High-Throughput Validation of Antibodies for Cancer Research

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Introduction

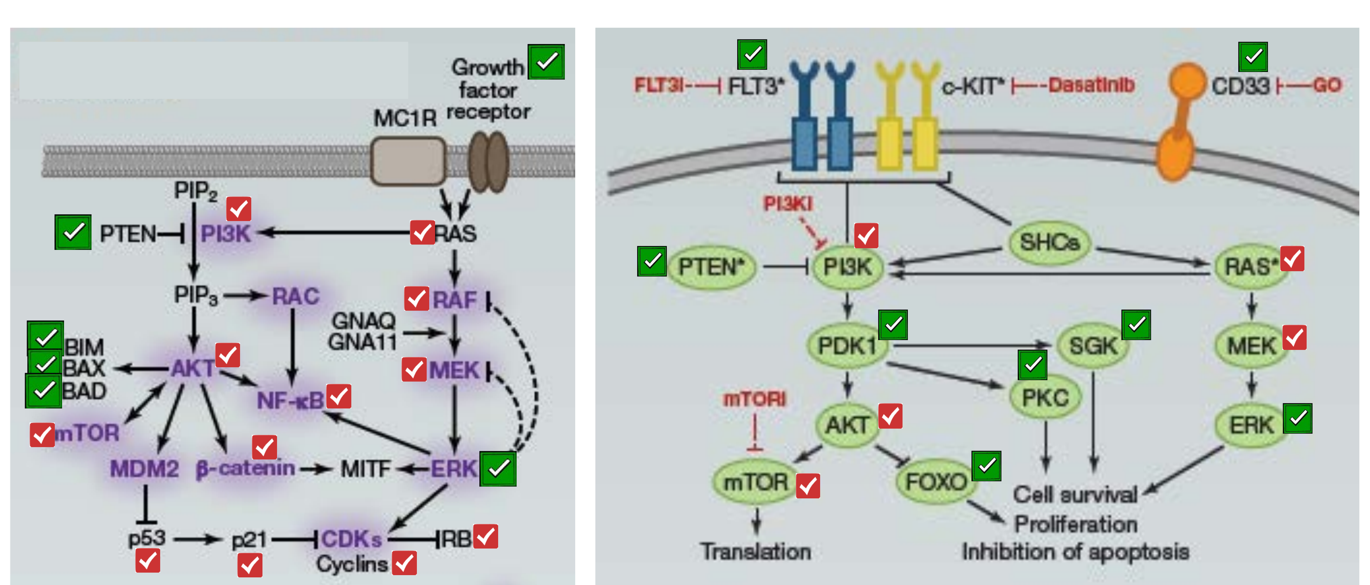
Antibodies are critical reagents for studying a wide range of cancers. Despite the need for high-quality antibodies that recognize proteins involved in cancers, only a small percentage of commercially available antibodies are well characterized and demonstrated to be of dependable quality.

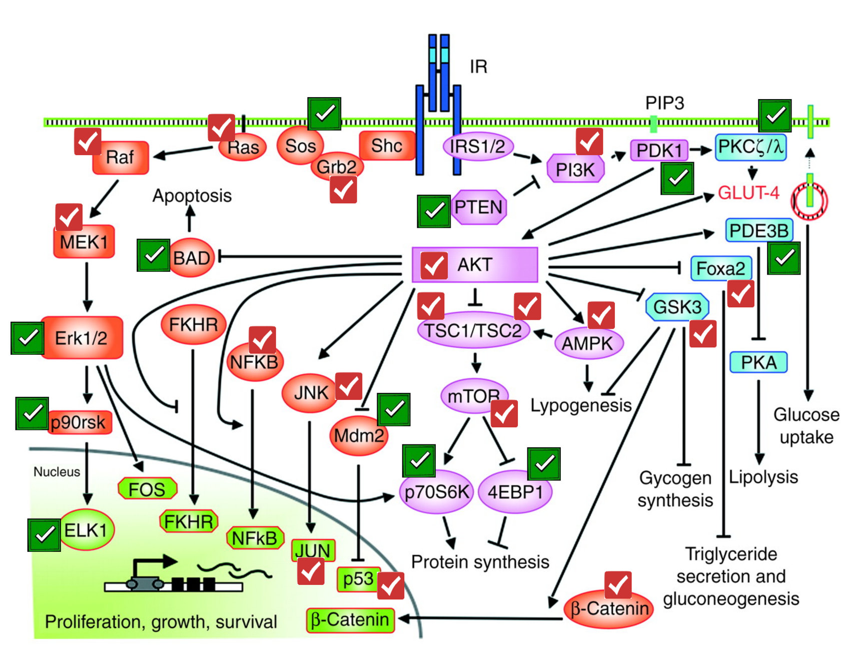
We have developed a robust and high-throughput process for validating antibodies to be used in western blotting for cancer research. This process is illustrated using several antibodies against key cancer-related targets, such as AKT1 and mTOR. Each antibody is screened against a panel of lysates from relevant tissue and cell samples expressing endogenous levels of the target protein. A rapid membrane transfer system coupled with a novel stain-free gel imaging technology ensures a high efficiency of protein transfer and allows us to assess sample quality. Antibody sensitivity and binding specificity are determined by quantitative data generated by our image analysis software.

This process has allowed us to identify high-performing antibodies for western blotting and other common antibody-based applications.

Common Cancer Pathways

Download Cancer Pathways Posters
https://www.bio-rad-antibodies.com/cancer-research-posters.html



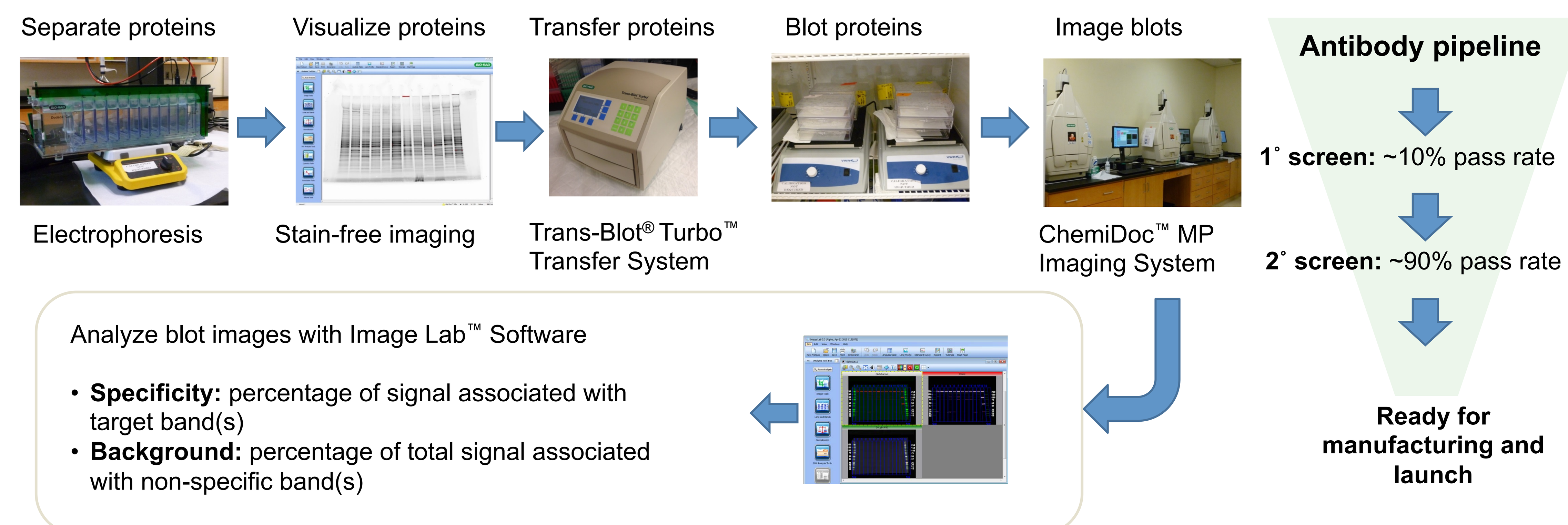


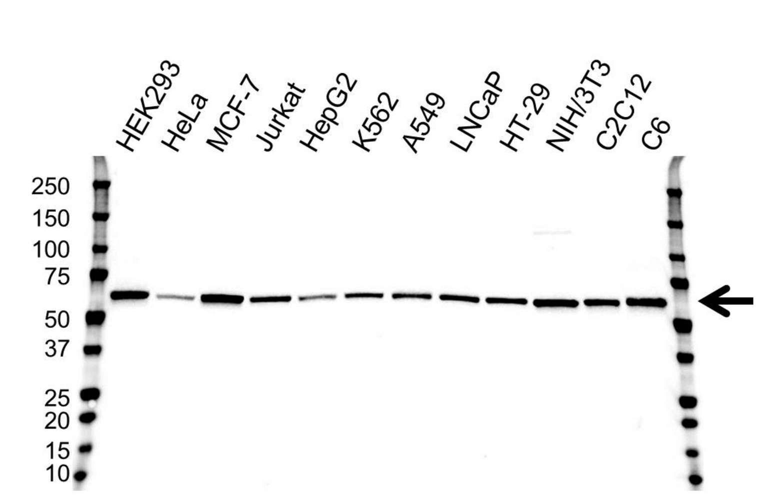
PrecisionAb Validated Western Blotting Antibodies

All others are available in the Bio-Rad antibody portfolio

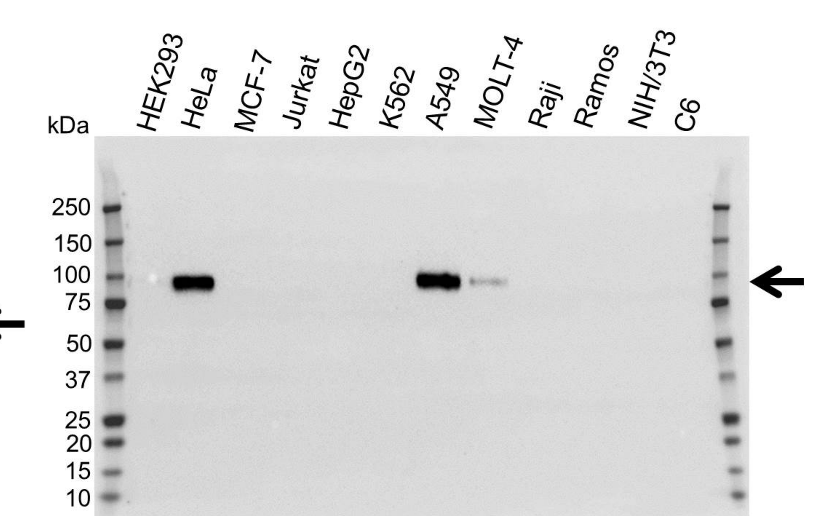
Belfiore A and Malaguarnera R (2011). Endocr Relat Cancer 18, R125–R147.

Antibody Validation Using the V3 Western Workflow™

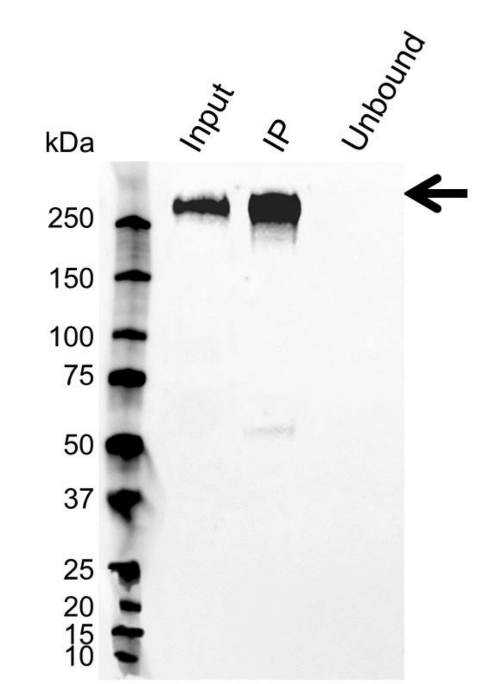


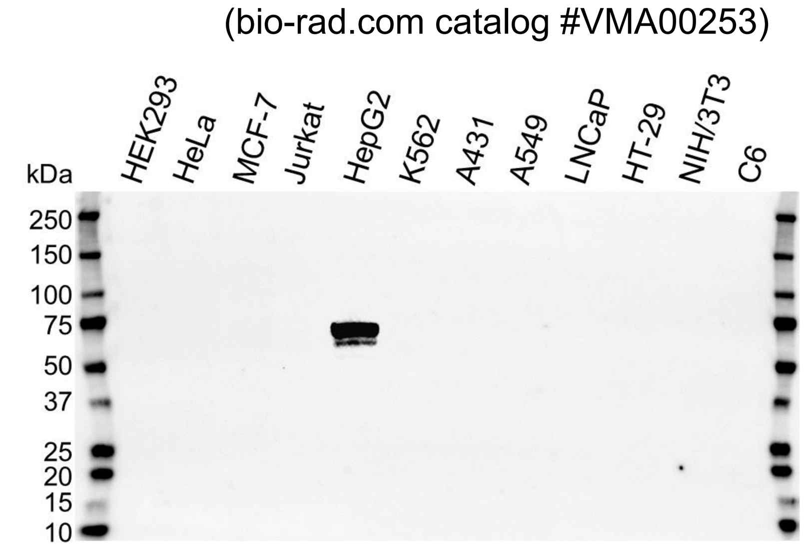


Mouse anti-human AKT1 (bio-rad.com catalog #VMA00253)

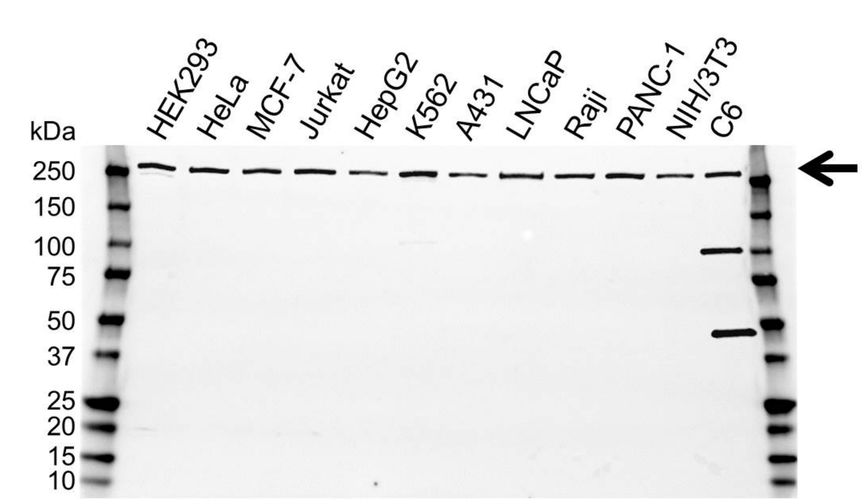


Mouse anti-human CD44 (bio-rad.com #VMA00034)





Mouse anti-human alpha fetoprotein (bio-rad.com #VMA00036)



Rabbit anti-human mTOR (bio-rad.com #VPA00174)

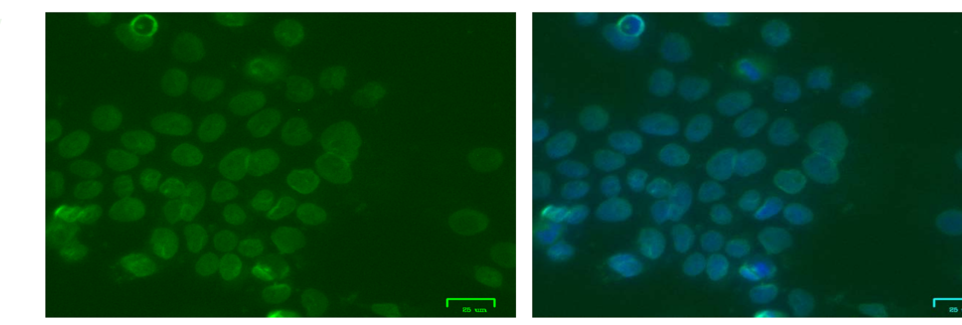
Immunoprecipitation. Human fibronectin was immunoprecipitated from LNCaP cells using a rabbit anti-human fibronectin antibody (bio-rad-antibodies.com #4470-2204). The resulting antibody-antigen complex was analyzed by western blotting using a sheep anti-human fibronectin antibody (bio-rad.com #VPA00045).

PrecisionAb™ Validated Western Blotting Antibodies

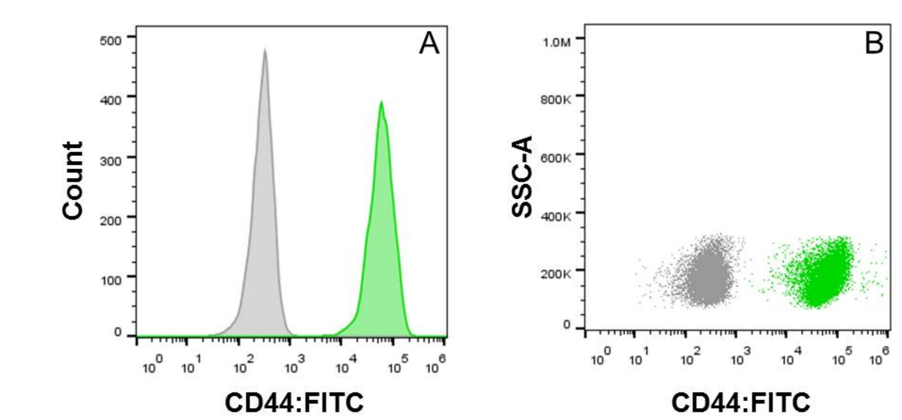
Highly Specific, Sensitive, and Dependable

- **Transparency** – full blot is shown (no cropping)
- **Stringent validation** – each antibody is tested on up to 12 biologically relevant lysates expressing endogenous levels of target (no overexpression)
- **Lot-to-lot consistency** for reproducible results

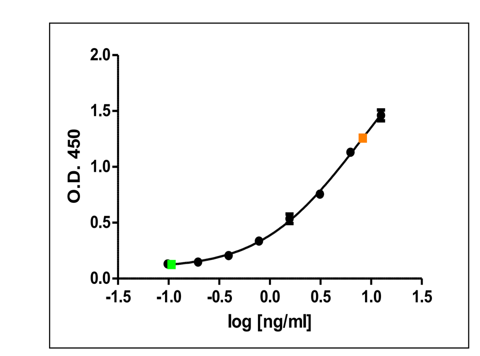
Beyond Western Blotting



Immunofluorescence. Fluorescence images of A431 cells stained with 2 µg/ml mouse anti-human XPO1 antibody (bio-rad.com #VMA00085) (green; left panel) and 20 µg/ml Hoechst 33258 (blue). The right panel is a merged image of green and blue fluorescence. (Images courtesy of Shanghai Antibody Center, Shanghai China)

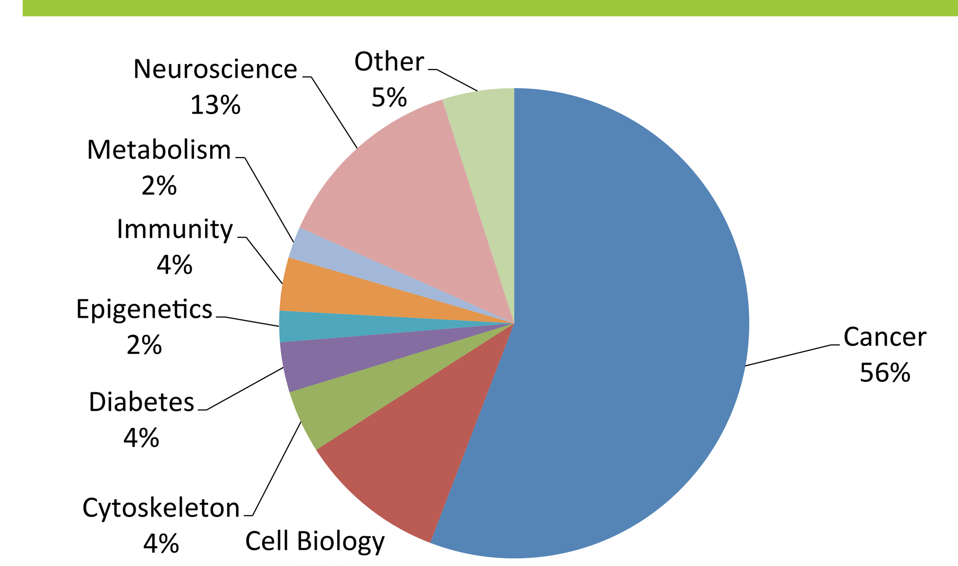


Flow cytometry. Detection of CD44 in normal human peripheral blood mononuclear cells (PBMCs) using the S3™ Cell Sorter. PBMCs were stained with a fluorescein isothiocyanate (FITC)–conjugated mouse anti-human CD44 antibody (bio-rad-antibodies.com #MCA89F) and lymphocytes were gated for analysis. Histogram overlay (**A**) and density plot (**B**) of CD44-stained cells (green) and unstained control cells (grey).



ELISA. Serial dilution of alpha fetoprotein (black squares), normal human serum (green), and plasma (orange) analyzed by sandwich ELISA. Antigen was captured with mouse anti-human alpha fetoprotein (bio-rad-antibodies.com #4520-5306) and detected with biotinylated mouse anti-human alpha fetoprotein (bio-rad-antibodies.com #0200-0562) and streptavidin-HRP.

PrecisionAb Portfolio



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